

TRAINING AN IMAGE CLASSIFIER ROBUST TO ADVERSARIAL ATTACKS

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<https://github.com/SaadH-077/adversarially-robust-image-classifier>

1 INTRODUCTION

Deep image classifiers are accurate on natural inputs yet collapse under imperceptible, worst-case perturbations; *adversarial training* hardens them by training on those perturbations [1]. We are given 50,000 images of shape $3 \times 32 \times 32$ with 9 classes, used directly as `float/255` $\in [0, 1]$ (no normalization), and must train a classifier robust to ℓ_∞ attacks with $\varepsilon = 8/255$. The architecture is restricted to `resnet18/34/50` from `torchvision` with only the final `fc` layer replaced to emit 9 logits; the ImageNet-style stem and all other layers are fixed, since we submit only the `state_dict` and it is loaded into the unmodified architecture. A submission is scored as $\text{score} = \frac{1}{2} A_{\text{clean}} + \frac{1}{2} A_{\text{rob}}$ and is rejected if $A_{\text{clean}} < 0.5$. The evaluation attack is hidden (“standard, strong, and diverse”), so the objective is *genuine* robustness rather than robustness overfit to one attack [8]. Our final submission reaches a public leaderboard score of **0.6430**.

2 APPROACH

The score weights clean and robust accuracy equally, so a strong submission must lift *both*. Our final model is a ResNet-34 trained with **Friendly–MART**: a misclassification-aware robust loss (MART) whose inner attack is the *friendly*, early-stopped PGD of FAT. We describe each component, then the design rationale.

Inner attack: PGD (Madry). For a clean input x with label y , the adversarial example comes from projected gradient ascent on the cross-entropy loss inside the ℓ_∞ ball, $x^{(t+1)} = \Pi_{\|x' - x\|_\infty \leq \varepsilon}(x^{(t)} + \alpha \text{sign} \nabla_{x^{(t)}} \mathcal{L}_{\text{CE}}(f(x^{(t)}), y))$, with $\varepsilon = 8/255$, step $\alpha = 2/255$, and a random start. This is the canonical robust-training attack [1] and our baseline (score 0.534).

Friendly early-stopping (FAT). Maximally adversarial examples make training conservative and depress clean accuracy. Friendly Adversarial Training [4] instead uses the *least*-adversarial example: PGD is run per-sample but *frozen* τ steps after the point is first misclassified ($f(x^{(t)}) \neq y$), and samples never misclassified within the step budget receive the full attack. The margin τ follows a curriculum, $\tau = \min(2, \lfloor \text{epoch}/20 \rfloor)$: early training ($\tau=0$) uses very friendly examples that build clean accuracy, and τ rises later to restore robustness. This is the single change that broke our accuracy/robustness plateau.

Robust loss: MART. On these examples we apply the MART objective [3], which augments a boosted cross-entropy on the adversarial input with a KL term that is up-weighted on *misclassified* examples:

$$\mathcal{L}_{\text{MART}} = \text{BCE}(f(x'), y) + \beta (1 - p_f(y | x)) \cdot \text{KL}(f(x) \| f(x')), \quad (1)$$

where x' is the friendly adversarial example, $p_f(y | x)$ is the clean predicted probability of the true class, BCE is cross-entropy plus a margin term that suppresses the most-likely wrong class, and $\beta = 3$. The KL/softmax terms are computed in `fp32` with logits clamped to $[-30, 30]$; without this, `fp16` underflow makes the term `nan` and the AMP gradient scaler silently skips every step.

Stabilisation, schedule, and hyperparameters. Adversarial training oscillates epoch-to-epoch, so we keep an exponential moving average (EMA, decay 0.999) of all weights and BatchNorm buffers [7] and use the EMA model. We train with SGD (lr 0.1, momentum 0.9, weight decay 5×10^{-4}) for 100 epochs at batch size 128, using a warmup then cosine schedule with SGDR warm restarts

[6] at epochs [40, 60, 80, 100]; the restarts specifically improve PGD robustness. The inner attack is FAT early-stopped PGD with up to 10 steps. We split 49k train / 1k held-out validation (seed 42), augment with RandomCrop(32, pad 4, reflect) and horizontal flips only, and select the checkpoint with the best $\frac{1}{2}A_{\text{clean}} + \frac{1}{2}A_{\text{PGD-7}}$ on the validation split.

Design rationale. The standard stack (PGD-AT $\rightarrow$ TRADES [2] $\rightarrow$ EMA $\rightarrow$ SGDR) plateaued near 0.595 on a flat frontier where $A_{\text{clean}} + A_{\text{rob}} \approx 1.19$; regularisers such as label smoothing and CutMix [10] only slid along it or gradient-masked. MART raised robustness but *gave up* clean accuracy, and an AutoAttack [9] audit showed part of that gain was a single-attack over-estimate, the obfuscated-gradients pitfall [8] (AutoAttack 0.445 vs. PGD 0.50). Two facts resolved the plateau: clean accuracy is not subject to the attack, so recovering it is a direct gain in the score, and our local PGD evaluation tracked the public leaderboard closely, giving a trustworthy proxy for selection. FAT recovers the clean accuracy MART discards and lifts both axes at once (Table 1); the remaining gains came from the clean axis, where ResNet-34 capacity, a friendlier curriculum ($\tau=2$), and more training data raised clean accuracy to $A_{\text{clean}}=0.770$ at $A_{\text{PGD-20}}=0.516$. A correctly implemented AWP [5] (proxy network, exact perturb/restore) raises robustness on the clean-limited ResNet-18, but on ResNet-34 it trades away the clean accuracy that is our edge, so the clean-axis variant is our final submission.

Table 1: Which lever moves which axis. $\nearrow$: improves the axis; $\approx$: roughly neutral; $\searrow$: hurts.

Lever	A_{clean}	A_{rob}
TRADES/MART robust loss	$\searrow$	$\nearrow$
EMA weight averaging	$\approx$	$\nearrow$
SGDR warm restarts	$\approx$	$\nearrow$
FAT friendly attack	$\nearrow$	$\approx$
ResNet-34 capacity	$\nearrow$	$\approx$
AWP weight perturbation	$\searrow$	$\nearrow$

Results. Table 2 shows the public leaderboard progression. The first ~ 0.06 came from the standard adversarial-training stack, which then plateaued; the decisive improvement was the FAT-MART method change (not a hyperparameter), and scaling the backbone to ResNet-34 added the final clean-accuracy margin. The final formula introduces no tuned thresholds or data-specific rules, limiting overfitting to the partial public ground truth.

Table 2: Public leaderboard score for each submitted configuration. All use $\varepsilon = 8/255$, EMA, and the SGDR schedule; loss/attack/backbone vary as shown.

Run	Configuration	Score
1	ResNet-18 PGD-AT, MultiStep LR	0.534
2	+ warmup $\rightarrow$ cosine, lr 0.1	0.556
3	ResNet-18 TRADES $\beta=3$ + EMA + SGDR	0.594
4	ResNet-34 TRADES + EMA + SGDR	0.595
5	ResNet-18 FAT-MART	0.624
6	ResNet-34 FAT-MART (τ cap 3)	0.628
7	ResNet-34 FAT-MART (τ cap 2, 49k data)	0.6430

3 CONCLUSION

Training on *friendly* (least-adversarial) examples rather than maximally adversarial ones lets a misclassification-aware loss recover the clean accuracy that aggressive adversarial training discards, lifting both axes to a leaderboard score of 0.6430 (clean 0.77, PGD-20 0.52). The stakes are practical: adversarial vulnerability blocks deployment of safety-critical systems such as autonomous driving and medical imaging, where an imperceptible perturbation flips a confident prediction. It also carries a caveat: robustness must be measured with strong, diverse attacks, since one attack over-stated ours in an obfuscated-gradients failure [8], caught by auditing with AutoAttack and PGD.

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